

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

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Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

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Introduction

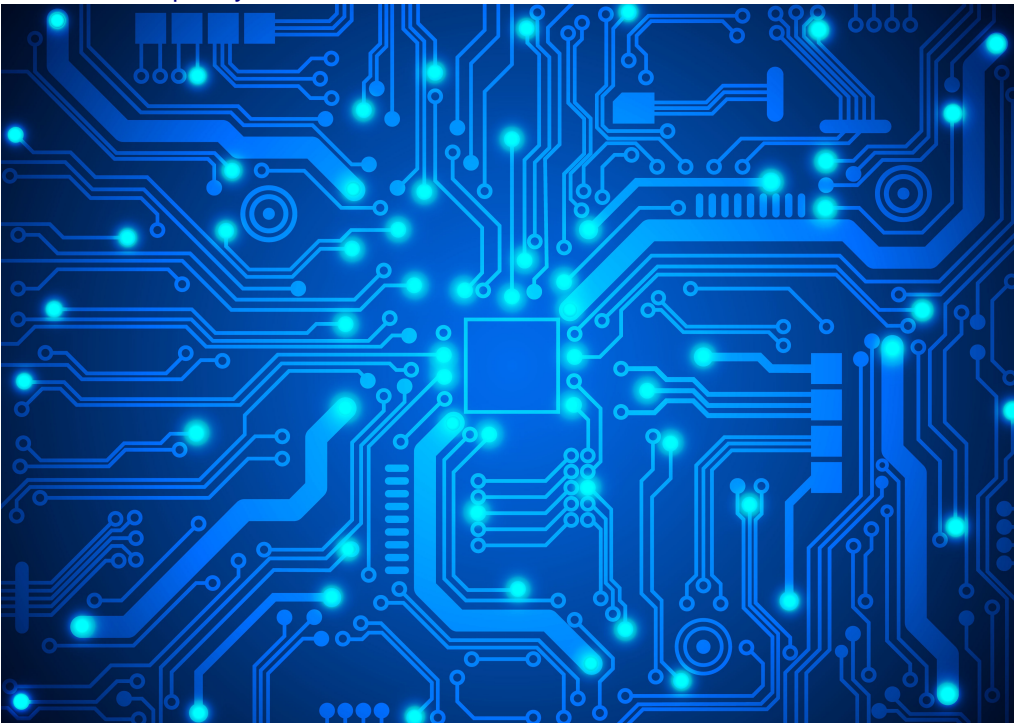


Introduction



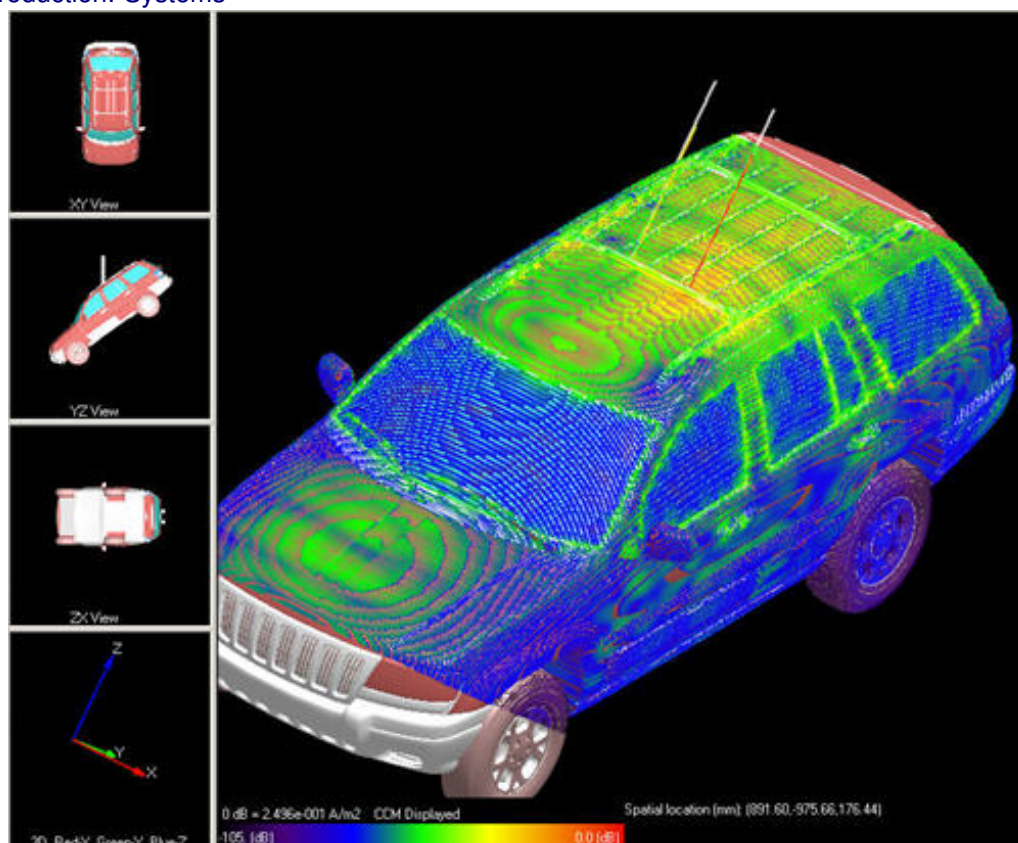
1.4

Introduction: Complexity



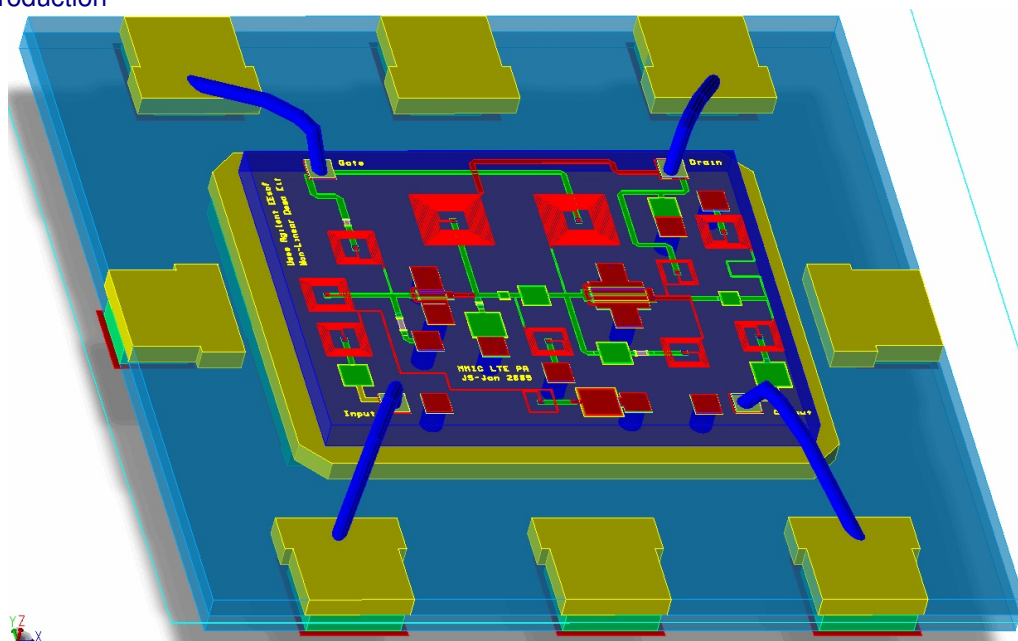
1.5

Introduction: Systems



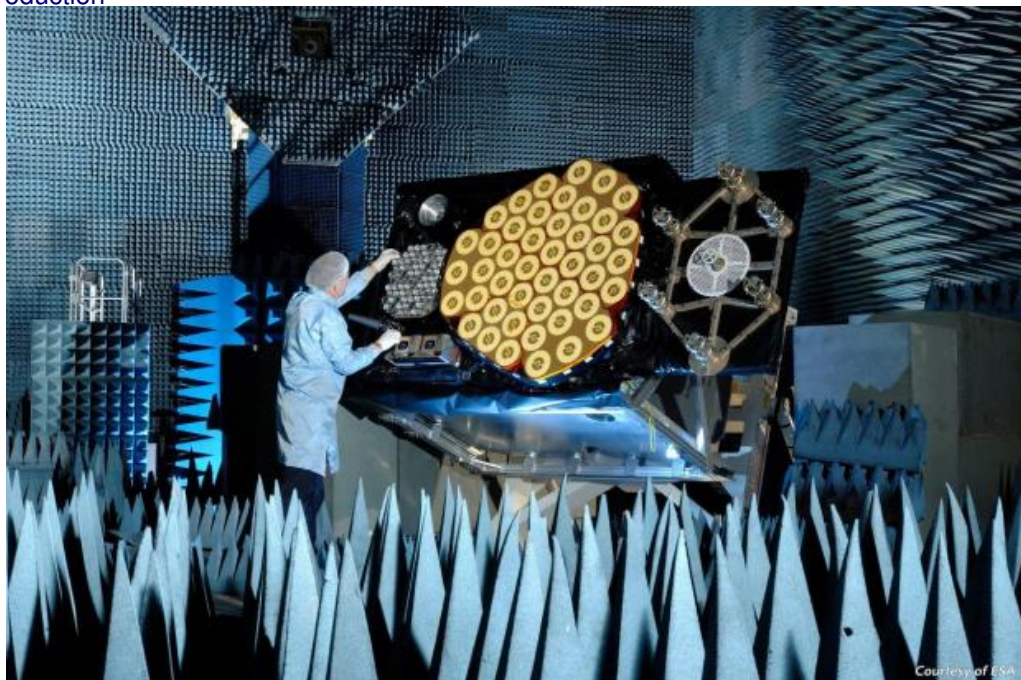
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 no lecture, pentacost break SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 live SR 00 014 (G.-Köhler-Allee 078)

1.9

Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

1.10

Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 1.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

1.11

Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

1.12

1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

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Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

1.14

WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

1.15

WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

1.17

Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

1.18

Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

1.19

Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: `bash` and `tcsh`

Start: `ads` after using `ssh` to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

1.20

Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

1.21

Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

1.22

The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Kesight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

29. june 2026

11.24

Please evaluate

Please give me feedback on the email you recieved this week !

There is a 20% quota!

Thanks for your feedback...

11.25

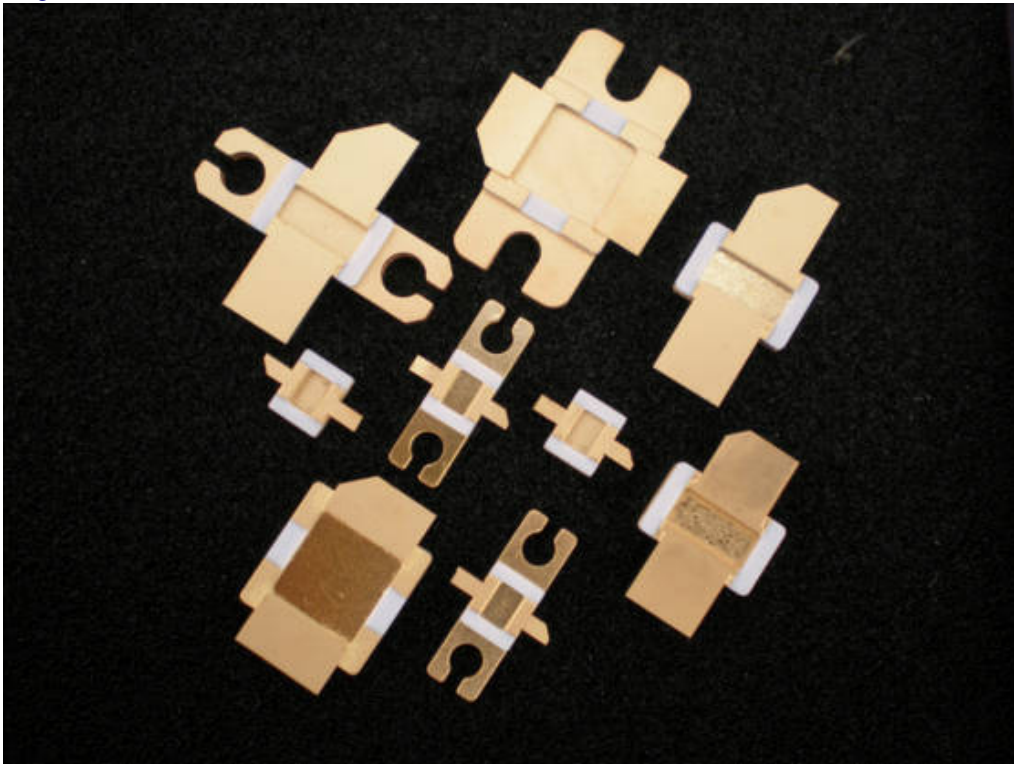
Part I

Part 3: Building Systems

2 Packages

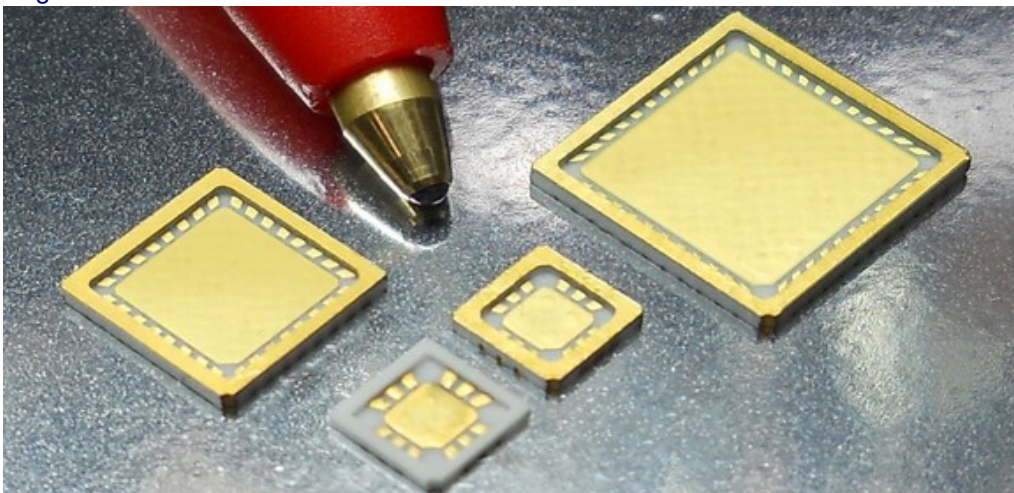
2.1 Package Design: Block 5A

Packages



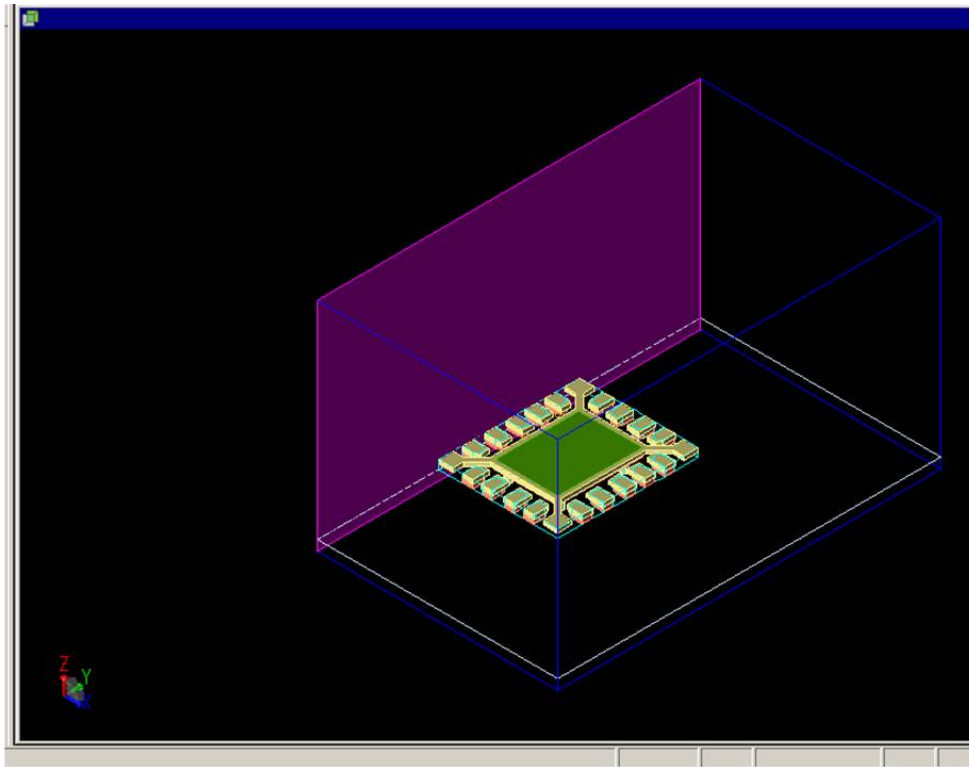
11.26

Packages



11.27

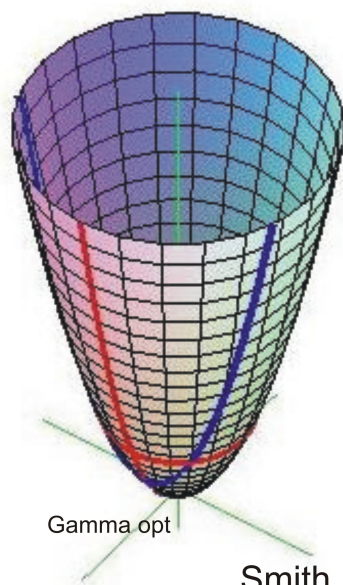
3D Simulation of an IC in a Package



11.28

The LNA matching

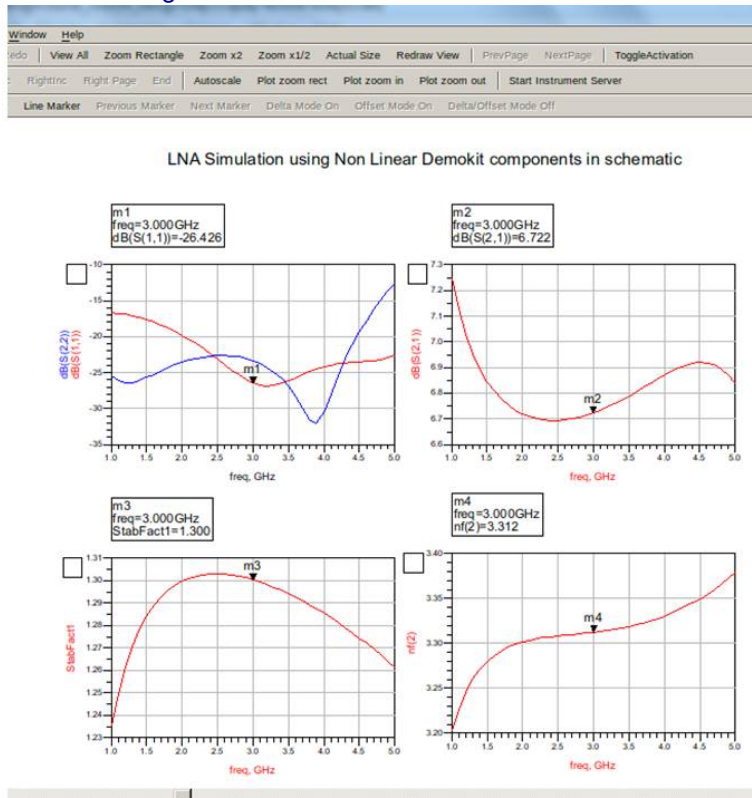
Fmin



Smith
Chart
for Gamma

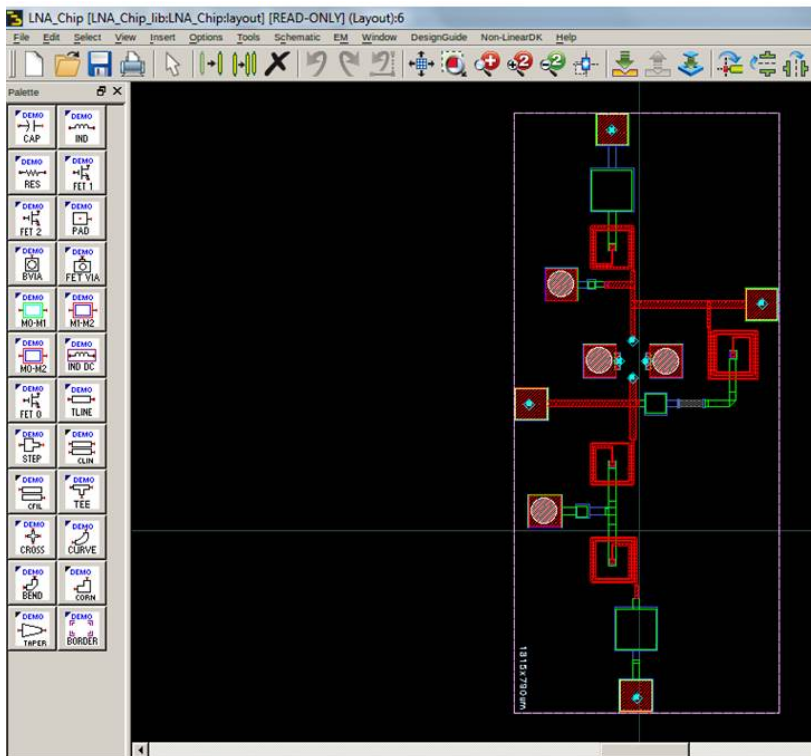
11.29

The LNA: Noise figure



11.30

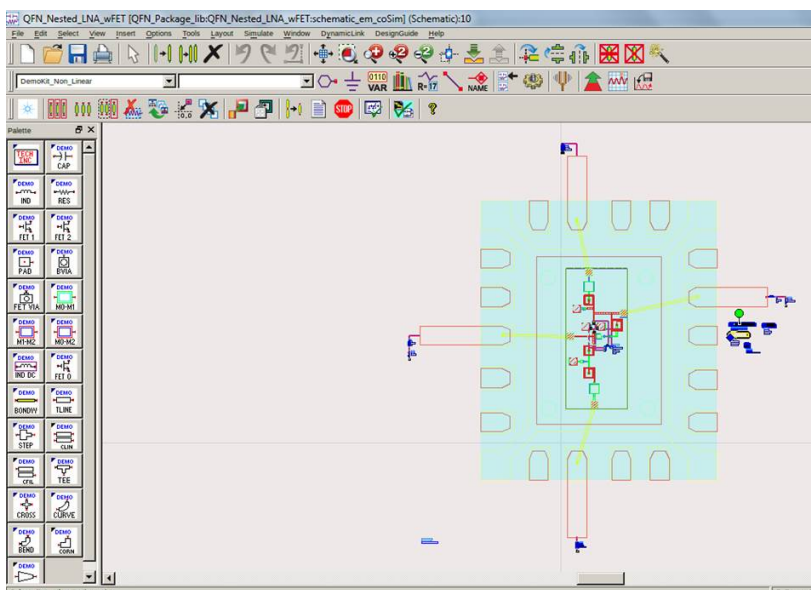
The LNA: Low-Noise Amplifier in Package



11.31

The package: Aspects of Cosimulation

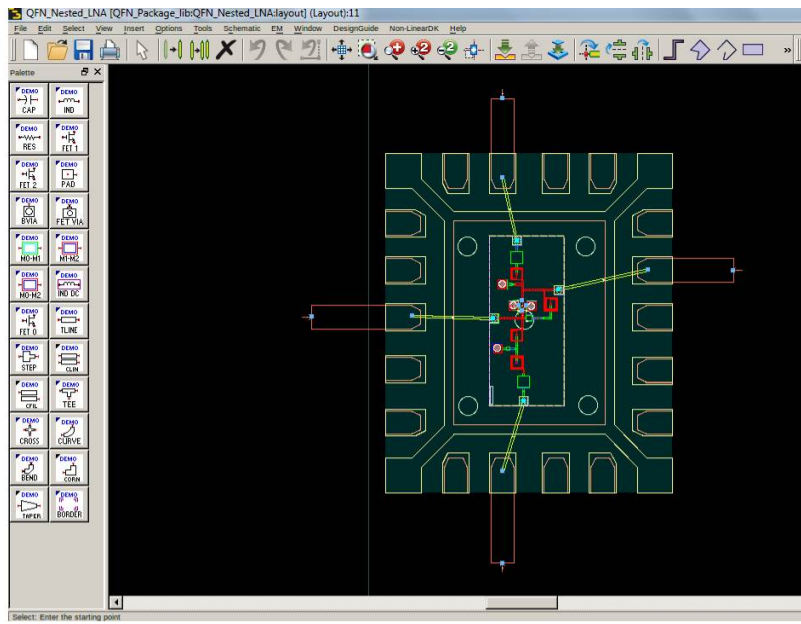
QFN: quad-flat no-leads package



11.32

The package: cosimulation

QFN: quad-flat no-leads package



2.2 Tasks

Tasks: Task 17 Block 5A

1. Explanation: This is about using multiple technologies in one simulator. So far we have looked at simulations of individual components or chips. The ADS tool can provide you with a multi-technology option, which allows you to simulate combinations of various technologies, represented by various design kits or libs. In this case the two technologies are one LNA MMIC and its package. It could also be various MMICs, filters, connecting substrates etc. The following, relatively simply example, will help you to understand, why this will be important for the final performance of a component or a system in package (SIP). Remember: at the end of the day, you will NOT be interested to have the best chips (ICs) on earth alone, but a component or better a function, which helps you to build a good overall system. Further this teaches you about tolerances: bondwires are not easily controlled in shape and length when manufactured over time or in greater numbers.

11.34

Tasks: Task 17: Packaging a MMIC in A QFN Package: block 5 A

2. Explanation: Open MTMFlowMMICChipQFNPackage in examples/Multi tech
3. Explanation Goto the schematic: LNAChip in Tech1
4. Task: Verify the stability for V_{dc} (src2)= 0.45, 1.05, 1.55, 2.05 V for the frequency range from 0.05 GHz to 4 GHz in steps of 10 MHz for the given gate bias by plotting. What do you observe for low frequencies below 1 GHz, i.e. in the 100 MHz range ? Why is this important (3 sentences)? What does happen, when the package gets instable (2 possibilities)?

11.35

Tasks: Task 17: Packaging a MMIC in A QFN Package

5. Task: Calculate the change of the S_{11} and S_{22} comparing a.) without (just schematic) and b.) with package for V_{dc} = 2.05 V, V_{dc} (gate)=-0.35 V. Discuss the changes of the matching when applying the package. Look at the frequencies to 0 Hz and also the relative changes... Name them in dB.
6. Task: Look at the package. Calculate the free space wavelength and its value free space wavelength/4 and related frequencies of the package in all three directions (x/y/z), which fit into the package. What can go wrong based on that with the system ? What does the package form BASED on the geometry, if you consider its dimension ? Think of a music instrument ! What happens with these wavelengths if you fill the package with a plastic mold with a dielectric constant of 2 ?
7. Task: If you see the pads as an MSL transmission line: which impedance do you attribute to it. Which impedance range can be realized with reasonable pads sizes ?

11.36

Tasks: Task 17: Packaging a MMIC in A QFN Package

8. Task: Add an inductor at the input of the device to 4 turns like the type L4 in the schematic in the connection from the source to the bondpad. This is to represent a bondwire added. Replot S_{11} . Compare S_{11} to the schematic situation without bondwire above. Add the same inductor at the output pad and replot S_{22} and compare to the original S_{22} . Again, discuss the relative change in dB and consider what this means to the packaged device.

11.37

Tasks: Task 17: Format Packaging a MMIC in A QFN Package

1. 17.4 Stability plot for four bias in one diagram and 3 sentences. One more sentence on what happen with the amplifier (2 possibilities)?
2. 17.5 Plot S_{11} and S_{22} in one diagram for a.) and b.) and three sentences.
3. 17.6 Four sentences.
4. 17.7 Give one example of the impedance and three sentences.
5. 17.8 One plot with both S_{11} and four sentences.

11.38